

PRD (Product Requirement Documentation)

Project Title: KisanShakti **Overview:** An integrated, mobile-first agronomic and financial ecosystem designed for smallholder farmers and hyperlocal buyers. The platform operates on a hybrid offline-online architecture to function in low-bandwidth rural areas.

Scope & Features

- **Weather & IoT Field Monitoring (Upaj):** Syncs regional weather API data with local ESP32 hardware via BLE/Hotspot to generate automated irrigation advisories.
- **AI-Based Crop Disease Detection (Upaj):** Offline Edge-AI (TFLite) computer vision pipeline to diagnose leaf diseases via smartphone camera.
- **Hyperlocal Commerce Engine (Mandi & Upaj):** Geospatial marketplace connecting farmers to nearby commercial buyers, using an ML-driven Smart Price benchmark based on Government Agmarknet data.
- **Farm Business Intelligence (Upaj):** Financial module automatically tracking incomes and expenses to calculate real-time ROI.

Success Criteria

- Disease detection inference executes completely offline via quantized INT8 model in under 2 seconds.
- Marketplace spatial queries return localized matches within a 200ms API response time.
- ESP32 BLE credential handover completes seamlessly without complex network configuration.

Out of Scope

- In-app payment gateways (transactions rely on local direct-contact and offline payments).
- Complex logistics tracking or delivery driver routing.
- Drone-based or satellite terrain mapping.